

logspace

Generate logarithmically spaced vector

Syntax

```
y = logspace(a,b)
y = logspace(a,b,n)
y = logspace(a,pi)
y = logspace(a,pi,n)
```

Description

`y = logspace(a,b)` generates a row vector `y` of 50 logarithmically spaced points between decades 10^a and 10^b . The `logspace` function is especially useful for creating frequency vectors. The function is the logarithmic equivalent of `linspace` and the `:` operator. [example](#)

`y = logspace(a,b,n)` generates `n` points between decades 10^a and 10^b . [example](#)

`y = logspace(a,pi)` generates 50 points between 10^a and π , which is useful in digital signal processing for creating logarithmically spaced frequencies in the interval $[10^a, \pi]$.

`y = logspace(a,pi,n)` generates `n` points between 10^a and π .

Examples

[collapse all](#)

Vector of Logarithmically Spaced Numbers

Create a vector of 50 logarithmically spaced points in the interval $[10^1, 10^5]$.

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```
y = logspace(1,5);
```

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Vector with Specified Number of Values

Create a vector of 7 logarithmically spaced points in the interval $[10^1, 10^5]$.

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```
y1 = logspace(1,5,7)
```

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```
y1 = 1×7  
105 ×
```

```
0.0001    0.0005    0.0022    0.0100    0.0464    0.2154    1.0000
```

Vector of Logarithmically Spaced Complex Numbers

Create a vector of complex numbers with 8 logarithmically spaced points between $10^{(1+2i)}$ and $10^{(5+5i)}$.

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```
y = logspace(1+2i,5+5i,8)
```

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```
y = 1×8 complex  
104 ×
```

```
-0.0001 - 0.0010i    0.0029 - 0.0024i    0.0133 + 0.0040i    0.0147 + 0.0497i    -0.1242 + 0.1479i
```

Input Arguments

[collapse all](#)

a — First bound

numeric scalar

First bound, specified as a numeric scalar. The `a` argument defines a bound of the interval over which `logspace` generates points. `a` can be real or complex, and the other bound, `b`, can be either larger or smaller than `a`. If `b` is smaller than `a`, then the vector contains descending values.

Data Types: `single` | `double`

Complex Number Support: Yes

b — Second bound

numeric scalar

Second bound, specified as a numeric scalar. The `b` argument defines a bound of the interval over which `logspace` generates points. `b` can be real or complex, and `b` can be either larger or smaller than the other bound, `a`. If `b` is smaller than `a`, then the vector contains descending values.

Data Types: `single` | `double`

Complex Number Support: Yes

n — Number of points

50 (default) | real numeric scalar

Number of points, specified as a real numeric scalar.

- If n is 1, `logspace` returns 10^b .
- If n is zero or negative, `logspace` returns an empty row vector.
- If n is not an integer, `logspace` rounds n down and returns `floor(n)` points.

Extended Capabilities

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Version History

Introduced before R2006a

See Also

[linspace](#) | [colon](#)

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